Spread Spectrum Codebooks for Narrowband Channels

Selecting faithful Walsh rows under bandwidth and noise limits

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Definitions

Channel The medium a signal travels through from transmitter to receiver (radio link, cable, ...). It blurs and adds noise.

Bandwidth The range of frequencies a channel can carry, in Hz. More bandwidth means the signal can change faster.

- **Narrowband:** a small range (a few Hz to a few hundred Hz). Only slowly changing signals get through.
- **Broadband:** a wide range. Fast signals and many users fit at once.

Spread spectrum Deliberately spreading a signal over more bandwidth than the data needs.

The problem

Why spread at all?

- Many users share band without scheduling
- Low probability of detection and interception
- Resistance to jammers

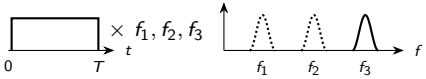
- Goal: Find a covert, jam-resistant way to broadcast over a narrowband channel, only a few Hz to a few hundred Hz wide.
- Spread spectrum normally buys covertness by using *more* bandwidth than the data needs.
- Here, narrowband channels have no spare bandwidth. Can we still spread?

Problem statement: Elmasry, *Using Spread Spectrum with Narrow Band Channels* (RTX).

Why classic spread spectrum doesn't fit

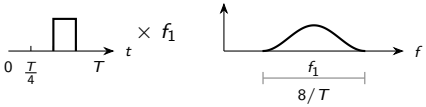
Frequency hopping

Square pulse, carrier hops between f_1, f_2, f_3 at random. Needs many bands: not available.



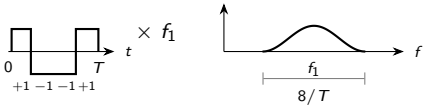
Time hopping

Send only in slot 3 of 4. The pulse is 4× shorter, so it uses 4× the bandwidth: not available.



Code-division multiple access

Use all of T, but multiply by a code $(+1, -1, -1, +1)$. Same spectrum as time hopping.



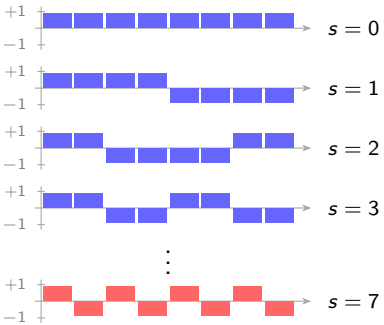
Hadamard matrices and Walsh ordering

± 1 matrices with orthogonal rows, $\mathbf{H}\mathbf{H}^{\mathsf{T}} = n\mathbf{I}_n$:

$$\mathbf{H}_{2n} = \begin{bmatrix} \mathbf{H} & \mathbf{H} \\ \mathbf{H} & -\mathbf{H} \end{bmatrix}$$

$$\mathbf{H}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad \mathbf{H}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

Sequency s = number of sign changes. Sorting rows by s gives the **Walsh matrix**.



The key idea

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Take an oversized Walsh matrix ($n = 512$).

Send every row through a bandlimited, noisy channel.

Keep the rows that still decode correctly: the faithful set.

Choose a small codebook (16 rows = 4 bits per word) from it.

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Faithful Set

Channel model

Faithful set size

Codebook Selection

Results

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Setup

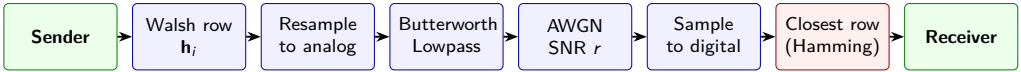
Faithful Set

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Channel model



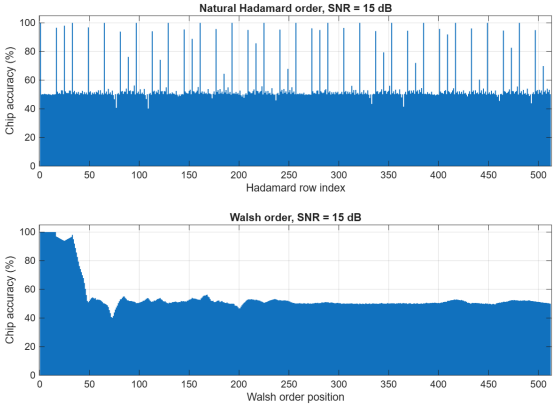
- Hadamard size $n = 512$; word rate $\omega_w = 10$ Hz; bandwidth $\omega_a = 512$ Hz
- SNR $r \in \{\infty, 3, 0, -3\}$ dB, 50 noisy runs each
- A row is faithful if it decodes to itself in every run
- From the faithful set, pick $2^w = 16$ codewords ($w = 4$ bits)

The faithful set

Definition

A Walsh row is **faithful** if it is decoded correctly in every noisy run. The faithful set $\mathcal{F}$ is what we pick codewords from.

- Same 512 rows, two orderings
- Natural order: survivors are scattered
- Walsh order: survivors group below the cutoff, $s \lesssim 40$



Chip accuracy by row, SNR $= 15$ dB, 20 runs, $n = 512$

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My methods

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Choosing 16 codewords

Faithful rows sorted by Walsh score $s_0 < \cdots < s_{n-1}$. Pick 16 that are as spread out as possible:

$$\max_{S \subset \mathcal{F},\ |S|=16} \ \min_{a \neq b \in S} |s_a - s_b|.$$

- **Inner min**: the closest pair in a candidate codebook S, i.e. its weakest link.
- **Outer max**: choose the 16 rows whose weakest link is as large as possible.
- Larger gaps mean codewords that are harder to confuse.
- Each selection algorithm is a different strategy for this max–min problem.

Greedy-First and Last

Both walk through the sorted faithful scores $s_1 < \cdots < s_m$ with a target spacing d^* (`optdis`) and pick 16 codewords.

Greedy First

- Next codeword: the **farthest** entry still at least d^* away
- Biased toward larger Walsh scores
- Gaps stay wide $\Rightarrow$ more error-resistant

Greedy Last

- Next codeword: the **farthest** entry within d^*
- Biased toward smaller Walsh scores
- Packs codewords tightly $\Rightarrow$ uses less of the band

Example: Scores $10, 15, 22, 28, 35, 42, 50, 58, 65, 72$, $d^* = 20$, pick 4



picks $10, 35, 58$ (then repeats 58), gaps $25, 23$



picks $10, 28, 42, 58$, gaps $18, 14, 16$

If no entry qualifies, both fall back to repeating the previous codeword.

Max-Gap Forward and Backward

Both binary-search the largest minimum gap d for which 4 rows fit, then keep that codebook.

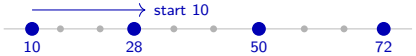
Max-Gap Forward

- Try a gap d (midpoint of the binary search).
- Start at the lowest score; scan up, keep a score if it is $\geq d$ above the last pick.
- 4 picks found: raise d. Too few: lower d.

Max-Gap Backward

- Try a gap d (midpoint of the binary search).
- Start at the highest score; scan down, keep a score if it is $\geq d$ below the last pick.
- 4 picks found: raise d. Too few: lower d.

Example: scores 10, 15, 22, 28, 35, 42, 50, 58, 65, 72, pick 4



picks 10, 28, 50, 72; gaps 18, 22, 22



picks 72, 50, 28, 10; gaps 22, 22, 18

Binary search on d: 31 fails, 15 fits, 23 fails, 19 fails, 17 fits, 18 fits, so $d^* = 18$ for both.

Greedy Farthest-Point and Quantile

Greedy Farthest-Point (GFP)

- Start at the lowest score.
- Each step, add the score with the max-min distance to the picks C:

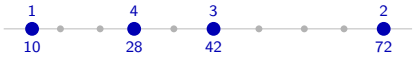
$$s^* = \arg\max_{s \notin C} \min_{c \in C} |s - c|$$

- Ties go to the lower score.

Quantile

- Split the scores into 4 equal-count bins.
- Target the bin-center quantile of each bin.
- Pick the unused score closest to each target.

Example: scores 10, 15, 22, 28, 35, 42, 50, 58, 65, 72, pick 4



Step	Picks so far	Added	Max-min
1	—	10	start
2	10	72	62
3	10, 72	42	30
4	10, 42, 72	28	14 (tie 58)

gaps 18, 14, 30



targets 13.75, 29.75, 48, 66.75;
picks 15, 28, 50, 65;
gaps 13, 22, 15

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Six-way comparison

Min distance

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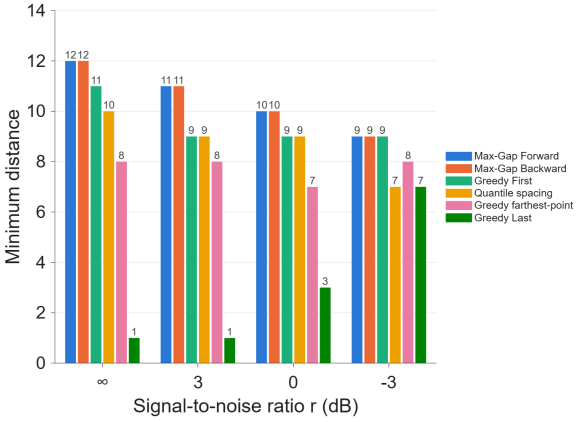
Results

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All six algorithms

Minimum distance, 16 codewords, $n = 512$

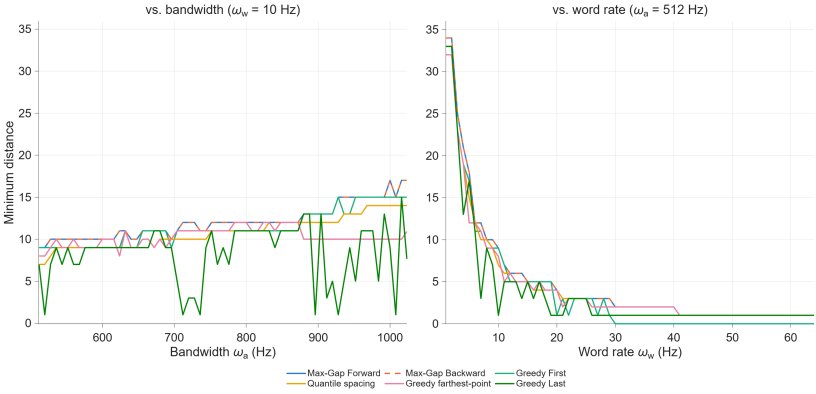
Algorithm	r (dB) ∞	3	0	-3
Max-Gap Forward	12	11	10	9
Max-Gap Backward	12	11	10	9
Greedy First	11	9	9	9
Quantile spacing	10	9	9	7
Greedy farthest-point	8	8	7	8
Greedy Last	1	1	3	7



- Max-Gap (forward and backward tie) has the widest spacing at every SNR.
- Greedy First is a close second and holds at 9 at all three noise levels.
- Greedy Last does worst: at $r = \infty$ and 3 dB two of its codewords are only 1 apart.

Minimum distance vs. bandwidth and word rate

Six Algorithms, 16 codewords, $n = 512$, $r = -3$ dB, 50 AWGN runs



- More bandwidth helps: Max-Gap grows from 9 to 17, with Greedy First close behind.
- Faster word rates make fewer rows fit in the band, so codewords are packed closer: doubling ω_w halves the min distance.

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Takeaways

Next steps

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Takeaways

- **CDMA can work on a narrowband channel** if we choose codes that fit it: slow-flipping Walsh rows survive the channel, and the faithful set $\mathcal{F}$ is exactly those rows.
- **Codebook choice is a spacing problem:** picking 16 codewords that are as far apart as possible within $\mathcal{F}$ sets how robust the system is to noise.
- **Max-Gap is the method to use:** it is provably optimal and wins at every SNR. Greedy First is a close, simpler alternative, and Greedy Last is unreliable.
- **The channel sets the limits:** min distance grows with bandwidth and shrinks as $1/\omega_w$, a direct trade between data rate and reliability.

Open questions and next steps

- Fading is a phenomenon where a signal's strength rises and falls over time as it passes through a medium. It can causes bursts, where one chip failing causes the next few to fail. When considering a model with fading, what level is acceptable to still produce sixteen faithful rows?
- Pre-processing: which algorithms are most successful in specific decibel ranges?

Thanks to George Elmasry, Dr. Pejman Sanaei, Lawan Wijayasooriya, and Dr. David Edwards.

Code: `https://github.com/a-ledra/RTX_MPI2026_Ledra`

Questions?